

Example 3: Simple Image Classification

Using Deep Network Designer and CNNs

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Agenda

1 Workflow Overview

2 Network Architecture

3 Understanding Dimensions (SSCB)

4 Training & Results

Workflow with Deep Network Designer

Step-by-step process:

- 1 **Design:** Drag & drop layers from the library.
- 2 **Connect:** Link layers sequentially.
- 3 **Analyze:** Verify dimension compatibility (S , T , C , B).
- 4 **Export:** Generate code or export network object (`net_1`) to workspace.

Predicted Class: 6



Predicted Class: 2



Predicted Class: 5



Figure 1: Deep Network Designer Interface.

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Input and Convolution Layers

1. Image Input Layer:

- Size: $28 \times 28 \times 1$ (Grayscale).
- Normalization: Zero-center (default).

2. Convolution 2D:

- **32 Filters** of size 3×3 .
- Feature extraction (edges, textures).
- Output depth becomes 32 chains.

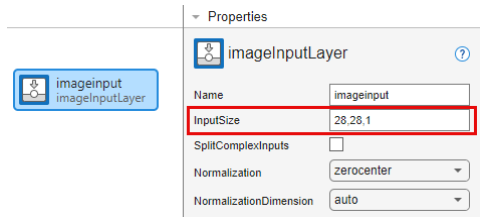


Figure 2: Input and Feature Extraction layers.

Processing and Activation

3. Batch Normalization:

- Stabilizes training by normalizing layer inputs.
- Allows for higher learning rates.

4. ReLU Layer:

- Introduces non-linearity: $f(x) = \max(0, x)$.
- Essential for learning complex patterns.

Classification Head

5. Fully Connected Layer:

- Output Size: **10** (Number of classes).
- Combines all local features for global decision.

6. Softmax Layer:

- Converts raw scores (logits) into probabilities summing to 1.

7. Classification Layer:

- Computes Cross-Entropy Loss during training.

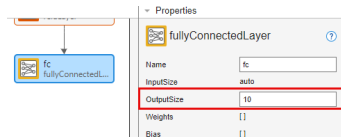


Figure 3: Full Architecture.

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Data Formats: S, T, C, B

MATLAB uses specific notation for tensors in the Network Analyzer.

Tag	Name	Span.	Description
S	Spatial	Espacial	Height and Width dimensions (e.g., 28×28). Reduced by pooling/convolution usually.
T	Time	Tiempo	Temporal dimension. For static images, $T = 1$.
C	Channel	Canal	Depth/Feature Maps. Input=1 (Gray), Output Conv=32.
B	Batch	Lote	Number of samples processed in parallel during training.

Analyzer View

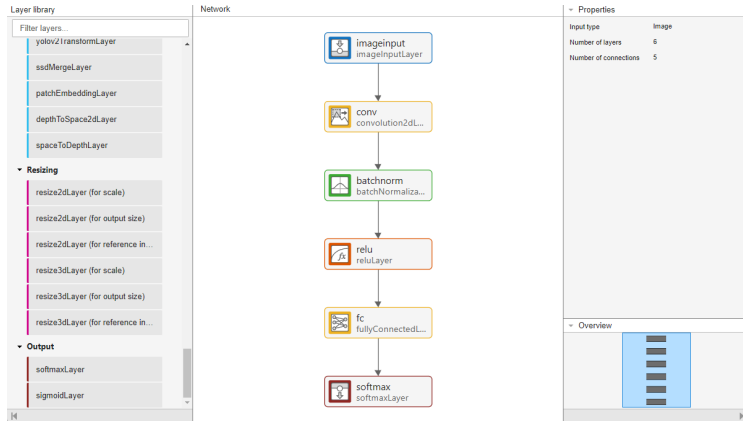


Figure 4: Network Analyzer Table showing Activations in SSCB format

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Training Progress

Training Config: sgdm, 4 Epochs, Accuracy Metric.

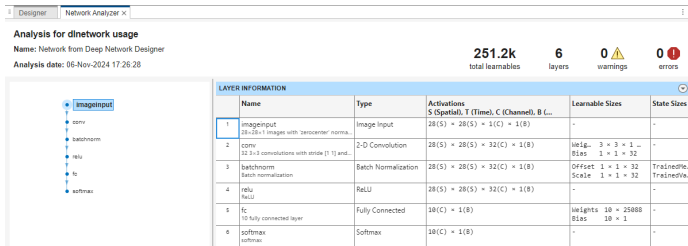


Figure 5: Training Progress Plot (Accuracy vs. Iteration).

Prediction Code

Using testnet or minibatchpredict:

```
1 % Train the network
2 net = trainnet(imdsTrain, net_1, "crossentropy", options);
3
4 % Evaluate standard accuracy
5 accuracy = testnet(net, imdsValidation, "accuracy");
6
7 % Visualizing Predictions
8 scores = minibatchpredict(net, imdsValidation);
9 YPred = scores2label(scores, classNames);
```

Validation Results

Example predictions on the validation set:

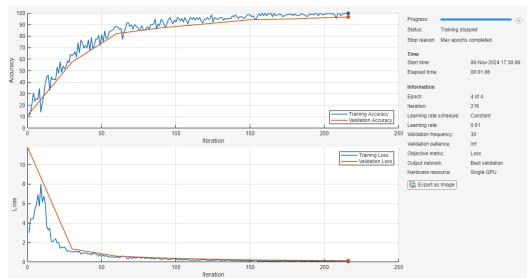


Figure 6: Prediction Grid.